

Towards Optimized Real-time Cybersickness Detection Framework Using Deep Learning for Standalone Virtual Reality Headsets

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ABSTRACT

Cybersickness (CS) is a major problem that is induced around 60-95% users due to immersive VR exposure, creates an obstacle for comfortable experiences. Deep learning models can predict CS, but require powerful resources, which is unsuitable for computationally constrained standalone VR (SVR) headsets. Using external computing resources can increase complexity. To address these gaps, we developed a framework for SVR devices where we trained deep learning models using physiological data, optimized the model, and deployed in the SVR headset to predict CS in real-time (during immersion) with minimum inference time. Our findings introduce future directions for real-time cybersickness prediction on SVR devices.

Keywords: cybersickness; cybersickness prediction; optimization; standalone virtual reality.

Index Terms: Human-centered computing—Human computer interaction—Interaction paradigms—Virtual reality

1 INTRODUCTION

Cybersickness is a type of motion sickness including headache, physical discomfort, dizziness, eye strain, heightened awareness of bodily sensations like stomach discomfort, etc., which is a significant barrier in the use of virtual reality (VR) technologies such as head-mounted displays [5]. Real-time CS mitigation depends on real-time CS detection and applying prompt mitigation techniques [3]. Traditional ways, such as simulator sickness questionnaires, take place after the immersion, which is not real-time as well as manual in nature, and the scores can be biased as they are subjective [5, 6]. Recently, a plethora of studies have been carried on to predict CS automatically using complex deep neural network (DNN) models and physiological data, which can produce promising result [4, 7]. However, those models were not optimized and not built for standalone devices, which require an external computation toolkit, causing an increase in complexity, network lag, and CS [8]. Also, they are not capable of making real-time CS predictions. To address this research gap, in this research, we focus on developing a framework to optimize the model and deploying in the standalone headset and to predict CS during VR immersion by considering the computational complexity and the inference time. Becoming motivated by the recent findings on SVR technology and efficient DNN model optimization for edge devices [1, 2], for this framework (Figure 1), we trained several DNN models, after evaluating their accuracy and inference time we selected the best one for further optimization to make the model efficient and deployable inside the SVR headsets. Our approach significantly reduces the model's complexity, which minimizes the inference time. As a result, using our framework, the

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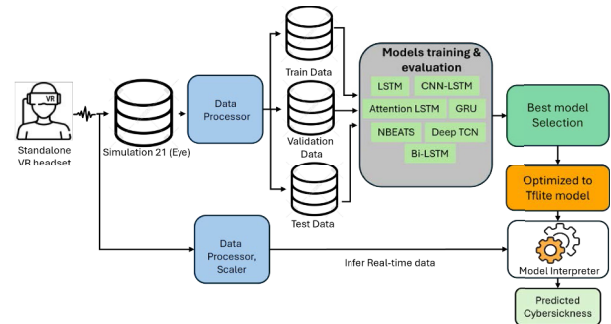
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Figure 1: An overview of our proposed framework for predicting real-time cybersickness from the best-optimized deep learning model.

Table 1: Error metrics (MSE, RMSE, MAE, and R^2 score) and parameters comparison of different models, showing NBEAST performs best.

Model	MSE	RMSE	MAE	R2 Score	Param
GRU	0.024	0.156	0.119	0.259	38113
LSTM	0.033	0.182	0.156	-0.006	91505
Atten.-LSTM	0.025	0.159	0.114	0.233	50253
Bi-LSTM	0.024	0.156	0.101	0.263	124657
CNN-LSTM	0.033	0.182	0.154	-0.003	42041
Deep TCN	0.03	0.173	0.126	0.095	171499
NBEATS	0.023	0.151	0.108	0.304	145283

SVR headset could run the simulation and predict CS fast motion scale (fms) value from the physiological data using the deployed DNN model during immersion. We validated our framework using the open source CS dataset [4] and Meta quest 3 as the SVR headset. In our study, our focus was eliminating external resource dependency (server & GPUs) and network lag, protecting data privacy, predicting CS during immersion, and utilizing the SVR in-built chipset.

2 METHODOLOGY

We employed a dataset provided by Islam et al. [4], which includes head-tracking, electrodermal activity, eye-tracking, heart rate data and corresponding fms values ranging from 0 to 10 as the ground truth from 27 participants observing different environments such as rollercoasters, beaches, roads, boats, and rooms. We only used eye tracking data as most SVR headsets have eye trackers. The eye tracking data has a total of 33 features (eye openness, pupil diameter, convergence distance, gaze origin, gaze orientation, normalized eye origin, and normalized spatial reference of gaze direction) of both left and right eyes. Developing this framework involves several steps, including dataset processing, model training, evaluation, best model selection and optimization, deploying the model inside the SVR headset, and evaluating the framework for predicting CS in

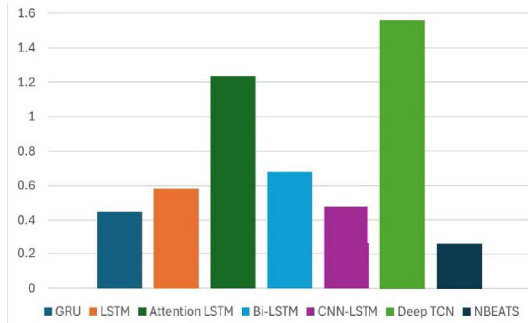


Figure 2: Comparison of inference time where NBEATS requires the lowest time to predict cybersickness.

real-time. We started the model training phase by processing the eye-tracking data by applying smoothing (moving average) and outlier-removing techniques (z-score analysis) and batch-making by using 28 time-steps of 33 features in each batch to hold the temporal characteristics of data. Then, split the batches of data into training (75%), test (10%), and validation (15%) sets based on the participants. We used Dell Precision 5820 computer and the same training and validation data sets to train and validate six DNN models (GRU, LSTM, Attention-LSTM, Bi-LSTM, CNN-LSTM, and NBEATS). The Tensorflow framework was used to build the neural basis expansion analysis for time series (NBEATS) model's architecture. After training, the same test set of data was employed for evaluating the models' performances. We carried out mean-squared-error (MSE), root-mean-squared-error (RMSE), mean-absolute-error (MAE), and R^2 score (Table 1), and we found that the NBEATS model architecture performed best, as the error metrics were the lowest. We also compared the inference time for all the trained models which is shown in Figure 2, where it shows that the inference time for NBEATS model architecture was lowest (0.062s) compared to others.

After model training and evaluation, we chose the NBEATS model as it performed best and required the lowest inference time. We employed TensorflowLite API to optimize the model by rendering it from the Tensorflow model into the Tensorflow lite model. In this model conversion, the bit length was pruned and quantized (reducing the precision of weights and activations). Consequently, the model size was reduced by maintaining the accuracy. After optimization, the model required only 0.042s as the inference time, which was around 84% reduced compared to the non-optimized model.

For deploying the optimized model in SVR headset we exported model and imported into the rollercoaster simulation run by Unity engine. This is an infinite-loop, noninteractive simulation which requires 57 seconds to complete each loop. To load the model, additional libraries, including the TensorflowLite interpreter were employed. In this study, we used Meta Quest 3, a standalone headset with no eye-tracker, so to mimic the ideal scenario, we imported the test set of data in the rollercoaster simulation from where data was fetched and fit to the model for CS prediction. The Meta Quest 3 headset is an Android operating system (OS)-based device and has the Snapdragon XR2 chipset (Octa-core Kryo CPU & Adreno 740 GPU). So, we rendered the simulation, including the model as an executable program compatible with the Android OS and Snapdragon chipset, and then deployed it in the headset.

3 RESULTS

In each phase of this study we evaluated the outcome in qualitative or quantitative analysis. From Table 1, we found that Nbeats performed better compared to LSTM-based model architecture. The MSE was

0.023, RMSE was 0.151, MAE was 0.108 and R^2 score was 0.304. These results indicate NBEATS as a strong cybersickness predictive model. Significantly lower inference time after optimization up to 84% made the model efficient to run in real-time with low-powered computationally constrained headsets. After deployment, the qual-



Figure 3: Screenshot of running the rollercoaster simulation, where predicting cybersickness (fms) is shown.

itative test was carried on by running the rollercoaster simulation in the headset itself. During the simulation running, the predicted cybersickness was visualized (Figure 3) to examine the framework.

4 CONCLUSION & FUTURE WORK

In this study, we trained several DNN models to find the best model for predicting cybersickness. We optimized the model to make it computationally efficient and minimize inference time for running on low-powered headsets. We deployed the simulation including model in the SVR headset and got the predicted CS during immersion. Though we got our preliminary result as expected but our study has the limitation in predicting users' real physiological data due to the lack of in-built eye-tracker. In future, the eye-tracker-contained standalone headset can be used to extend this framework for predicting cybersickness based on the user's eye-tracking data during immersive moments.

REFERENCES

- [1] Y. Deng. Deep learning on mobile devices: a review. In *Mobile Multi-media/Image Processing, Security, and Applications 2019*, vol. 10993, pp. 52–66. SPIE, 2019.
- [2] M. J. Islam and R. Islam. Towards optimized cybersickness prediction for computationally constrained standalone virtual reality devices. In *2024 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pp. 1003–1004. IEEE, 2024.
- [3] R. Islam, S. Ang, and J. Quarles. Cybersense: A closed-loop framework to detect cybersickness severity and adaptively apply reduction techniques. In *2021 IEEE Conference on virtual reality and 3d user interfaces abstracts and workshops (VRW)*, pp. 148–155. IEEE, 2021.
- [4] R. Islam, K. Desai, and J. Quarles. Cybersickness prediction from integrated hmd's sensors: A multimodal deep fusion approach using eye-tracking and head-tracking data. In *2021 IEEE international symposium on mixed and augmented reality (ISMAR)*, pp. 31–40. IEEE, 2021.
- [5] R. S. Kennedy, N. E. Lane, K. S. Berbaum, and M. G. Lillenthal. Simulator sickness questionnaire: An enhanced method for quantifying simulator sickness. *The international journal of aviation psychology*, 3(3):203–220, 1993.
- [6] H. K. Kim, J. Park, Y. Choi, and M. Choe. Virtual reality sickness questionnaire (vrsq): Motion sickness measurement index in a virtual reality environment. *Applied ergonomics*, 69:66–73, 2018.
- [7] J. Kim, W. Kim, H. Oh, S. Lee, and S. Lee. A deep cybersickness predictor based on brain signal analysis for virtual reality contents. In *Proceedings of the IEEE/CVF international conference on computer vision*, pp. 10580–10589, 2019.
- [8] J.-P. Stauffert, F. Niebling, and M. E. Latoschik. Latency and cybersickness: Impact, causes, and measures. a review. *Frontiers in Virtual Reality*, 1:582204, 2020.